

1. Executive Role Overview & Scope

In **APML Connect Pro** (Enterprise EMR & Clinical Management Suite), the **Data Administrator (Database & Health Data Administrator)** plays a pivotal role in ensuring complete data integrity, system security, HIPAA/GDPR regulatory compliance, and seamless peer-to-peer data synchronization across clinical workstations (Reception Desk, Doctor Cabinet, Pharmacy Counter, and Mobile Apps).

The Data Administrator maintains strict oversight over database schemas (Prisma ORM & PostgreSQL/SQLite), encryption keys, user access privileges, audit logs, disaster recovery protocols, and offline-first mesh synchronization performance (`meshSync`).

2. Key Core Responsibilities Matrix

Database Architecture & Schema Management	- Manage Prisma ORM schema migrations & model relationships. - Execute database DDL/DML updates without service downtime. - Maintain indexes, foreign keys, and query performance execution plans.	schema.prisma PostgreSQL / SQLite Prisma Client
Offline Mesh Sync Oversight (`meshSync`)	- Monitor WebSocket real-time data replication streams across workstations. - Resolve vector clock & timestamp merge conflicts upon node reconnect. - Validate client LocalStorage cache policies & queue token persistence.	WebSocket Gateway Local Cache Stores P2P Sync Engine
Security & RBAC Enforcement	- Provision & audit user roles (Super Admin, Hospital Admin, Doctor, Receptionist, Pharmacist, Lab Tech). - Configure multi-tenant isolation rules & JWT token expiry. - Oversee AES-256 envelope encryption for sensitive PHI columns.	RBAC Access Matrix AWS KMS / Vault JWT Auth Middleware
Regulatory Audit Logging (HIPAA)	- Enforce immutable append-only logging for EMR record access. - Perform weekly security access reviews & anomalous query detection. - Maintain audit log storage for statutory retention periods.	AuditLog Table HIPAA § 164.312(b) Security Logs
Backup & Disaster Recovery (DR)	- Configure automated Point-In-Time-Recovery (PITR) database snapshots. - Oversee offsite multi-AZ encrypted backups for clinical files & PDFs. - Conduct periodic DR failover drills & database restore validations.	PostgreSQL PITR AWS S3 Storage Disaster Recovery Plan
Data Standardization & Master Records	- Standardize ICD-10 diagnostic codes & pharmacy drug inventories. - Execute Master Patient Index (MPI) deduplication & record merging. - Manage legacy EMR data imports, ETL pipelines, and executive reporting.	MPI System Pharmacy Inventory ETL Scripts

3. Data Administrator Privilege & Permission Boundaries

- Authorized Privileges:**
- Full Schema & DDL Control:** Execute Prisma migrations, alter table structures, and create indexes.
 - User Credential & Role Management:** Assign role permissions, configure 2FA/TOTP parameters, and revoke compromised accounts.
 - Audit Log Inspection:** View system-wide access logs, audit trails, and data transfer metrics.
 - Database Restoration & Maintenance:** Trigger manual backups, perform table vacuums, and initiate failovers.
- Strict Compliance Boundaries (HIPAA Privacy Rule):**
- No Unauthorized PHI Decryption:** Data Administrators cannot decrypt raw patient clinical diagnosis notes or medical files without explicit audit authorization.
 - Immutable Audit Restrictions:** Data Administrators cannot edit, delete, or overwrite entries in the AuditLog table.

4. Daily & Weekly Operational Workflows

■ Daily Operations

1. Check DB health, memory & connection pools.
2. Verify meshSync WebSocket node connectivity.
3. Review failed sync queues & retry logs.
4. Ensure daily PITR snapshot execution.

■ Weekly Operations

1. Perform audit log anomaly analysis.
2. Execute database index defragmentation.
3. Verify S3 backup integrity & encryption keys.
4. Deduplicate Master Patient Index (MPI).

■ Emergency Operations

1. Initiate instant database failover on server crash.
2. Reconcile desynchronized peer nodes.
3. Lock down accounts on suspicious access.
4. Execute emergency restore from PITR point.